

A Practical Guide to Recurrent neural network

TOPIC -- RECURRENT NEURAL NETWORK

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

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Fully recurrent neural networks (FRNN) connect the outputs of all neurons to the inputs of all neurons. In other words, it is a fully connected network. This is the most general neural network topology, because all other topologies can be represented by setting some connection weights to zero to simulate the lack of connections between those neurons.

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Teacher forcing An RNN can be trained into a conditionally generative model of sequences, aka autoregression. Concretely, let us consider the problem of machine translation, that is, given a sequence $(x_1, x_2, \dots, x_n)$ of English words, the model is to produce a sequence $(y_1, \dots, y_m)$ of French words. It is to be solved by a seq2seq model. Now, during training, the encoder half of the model would first ingest $(x_1, x_2, \dots, x_n)$, then the decoder half would start generating a sequence $(\hat{y}_1, \hat{y}_2, \dots, \hat{y}_l)$. The problem is that if the model makes a mistake early on, say at $\hat{y}_2$, then subsequent tokens are likely to also be mistakes. This makes it inefficient for the model to obtain a learning signal, since the model would mostly learn to shift $\hat{y}_2$ towards y_2 , but not the others. Teacher forcing makes it so that the decoder uses the correct output sequence for generating the next entry in the sequence. So for example, it would see $(y_1, \dots, y_k)$ in order to generate $\hat{y}_{k+1}$.

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The Hopfield network is an RNN in which all connections across layers are equally sized. It requires stationary inputs and is thus not a general RNN, as it does not process sequences of patterns. However, it guarantees that it will

converge. If the connections are trained using Hebbian learning, then the Hopfield network can perform as robust content-addressable memory, resistant to connection alteration.

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Long short-term memory (LSTM) is the most widely used RNN architecture. It was designed to solve the vanishing gradient problem. LSTM is normally augmented by recurrent gates called "forget gates". LSTM prevents backpropagated errors from vanishing or exploding. Instead, errors can flow backward through unlimited numbers of virtual layers unfolded in space. That is, LSTM can learn tasks that require memories of events that happened thousands or even millions of discrete time steps earlier. Problem-specific LSTM-like topologies can be evolved. LSTM works even given long delays between significant events and can handle signals that mix low and high-frequency components. Many applications use stacks of LSTMs, for which it is called "deep LSTM". LSTM can learn to recognize context-sensitive languages unlike previous models based on hidden Markov models (HMM) and similar concepts.

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Introduced by Bart Kosko, a bidirectional associative memory (BAM) network is a variant of a Hopfield network that stores associative data as a vector. The bidirectionality comes from passing information through a matrix and its transpose. Typically, bipolar encoding is preferred to binary encoding of the associative pairs. Recently, stochastic BAM models using Markov stepping are optimized for increased network stability and relevance to real-world applications. A BAM network has two layers, either of which can be driven as an input to recall an association and produce an output on the other layer.

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A recursive neural network is created by applying the same set of weights recursively over a differentiable graph-like structure by traversing the structure in topological order. Such networks are typically also trained by the reverse mode of automatic differentiation. They can process distributed representations of structure, such as logical terms. A special case of recursive neural networks is the RNN whose structure corresponds to a linear chain. Recursive neural networks have been applied to natural language processing. The recursive neural tensor network uses a tensor-based composition function for all nodes in the tree.

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An RNN-based model can be factored into two parts: configuration and architecture. Multiple RNNs can be combined in a data flow, and the data flow itself is the configuration. Each RNN itself may have any architecture, including LSTM, GRU, etc.

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Neural history compressor The neural history compressor is an unsupervised stack of RNNs. At the input level, it learns to predict its next input from the previous inputs. Only unpredictable inputs of some RNN in the hierarchy become inputs to the next higher level RNN, which therefore recomputes its internal state only rarely. Each higher level RNN thus studies a compressed representation of the information in the RNN below. This is done such that the input sequence can be precisely reconstructed from the representation at the highest level. The system effectively minimizes the description length or the negative logarithm of the probability of the data. Given a lot of learnable predictability in the incoming data sequence, the highest level RNN can use supervised learning to easily classify even deep sequences with long intervals between important events. It is possible to distill the RNN hierarchy into two RNNs: the "conscious" chunker (higher level) and the "subconscious" automatizer (lower level). Once the chunker has learned to predict and compress inputs that are unpredictable by the automatizer, then the automatizer can be forced in the next learning phase to predict or imitate through additional units the hidden units of the more slowly changing chunker. This makes it easy for the automatizer to learn appropriate, rarely changing memories across long intervals. In turn, this helps the automatizer to make many of its once unpredictable inputs predictable, such that the chunker can focus on the remaining unpredictable events. A generative model partially overcame the vanishing gradient problem of automatic differentiation or backpropagation in neural networks in 1992. In 1993, such a system solved a "Very Deep Learning" task that required more than 1000 subsequent layers in an RNN unfolded in time.